

Innlevering 2 Mas117

oppgave 2.1:

$$20 [\text{klasserom}] \quad 110 [\text{W/lyster}]$$

$$200 [\text{kontor}]$$

$$2 [\text{lyster/kontor}] \quad 1,5 [\text{kr/kWh}]$$

$$18 [\text{lyster/klasserom}] \quad \text{ans} [\text{kr/år}]$$

dager
åpent

$$1 [\text{år}] = 240 [\text{dager}]$$

$$1 [\text{dager}] = 24 [\text{h}]$$

$$1 [\text{W}] = 1 [\text{J/s}]$$

$$1 [\text{kWh}] = 1000 [\text{W}] \cdot 1 [\text{h}]$$

$$1 [\text{h}] = 60 [\text{min}]$$

$$1 [\text{min}] = 60 [\text{s}]$$

[kW]

$$\dot{E} = 20 \cdot 18 + 200 \cdot 2 = 760 [\text{J/s}] = 0,76 [\text{W/s}]$$

$$360 + 400$$

$$(24 \cdot 240) [\text{h/år}]$$

$$E = \int_0^{\dot{E}} dt = 0,76 [\text{kW}] \cdot (24 \cdot 240) [\text{h/år}] = 4377,6 [\text{kWh/år}]$$

$$1,5 [\text{kr/kWh}] \cdot 4377,6 [\text{kWh/år}] = 6566,4 [\text{kr/år}]$$

oppg. 2.2:
a)

$$S = 20 \cdot 10^0$$

$$\dot{V} = 60 \cdot 10^{-3} [\text{m}^3/\text{s}]$$

$$\dot{E}_s = 17,5 \cdot 10^3 [\text{VA}]$$

$$\rho = 977 [\text{kg}/\text{m}^3]$$

$$\dot{m} = \dot{V} \rho$$

$$[J] = \left[\text{m} \cdot \frac{\text{m}^3}{\text{s}^2} \cdot \text{kg} \right] \rightarrow E = Fs = m a s, a = g = 9,81 [\text{m}/\text{s}^2]$$

$$\dot{E} = \dot{F} s = \dot{m} g s = \dot{V} \rho g s = 60 \cdot 10^3 [\text{m}^3/\text{s}] \cdot 977 [\text{kg}/\text{m}^3] \cdot 9,81 [\text{m}/\text{s}^2] \cdot 20 [\text{m}] = 11501,244$$

$$\left[\text{kg} \frac{\text{m}^2}{\text{s}^3} \right] \rightarrow \left[\frac{\text{J}}{\text{s}} \right] \rightarrow [\text{W}]$$

$$\frac{11501,244 [\cancel{\text{W}}]}{17,5 \cdot 10^3 [\cancel{\text{W}}]} = 0,657213$$

b)

$$p = \rho g h = 977 \cdot 9,81 \cdot 20 = 191687,4 [\text{Pa}]$$

$$\approx 192 [\text{kPa}]$$

$$p_{\text{pumpe}} > p$$

$$p_{\text{pumpe}} > 192 [\text{kPa}]$$

oppg. 1.3:

a)

anpartar elektriske hestekrefter
der vi direkte skal finne tilført effekt [W]
på en elektrisk motor.

$$\dot{E}_m = 75 [\text{kW}] \cdot 0,746 [\text{kWh/kW}] \cdot 0,75 [\% \text{Lust}] = 41,9625 [\text{kW}]$$

$$\dot{E}_{EL1} = \dot{E}_m \cdot \eta_1 = 41,9625 \cdot \frac{1}{0,91} = 46,11263 [\text{kW}]$$

$$\dot{E}_{EL2} = \dot{E}_m \cdot \eta_2 = 41,9625 \cdot \frac{1}{0,954} = 43,98549 [\text{kW}]$$

$$\Delta \dot{E}_{EL} = \dot{E}_{EL2} - \dot{E}_{EL1} \approx -2,12 [\text{W}]$$

≈ 2 watt mindre enn før motor

$$- 2,12 \cdot 10^{-3} [\text{kW}] \cdot 4368 [\text{h/år}]$$

$$\approx -9,2898 [\text{kWh/år}]$$

b) $\cdot 1,5 \text{ kr/kWh}$

$$\approx -13,9347 [\text{kr/år}]$$

↑

hvor mye endring i pengeforbruk per år
er negativt, så du sparer penger

du sparer ca. 14 [kr] i året på å bytte motor

oppg. 2.4:

a) $5,5 \cdot 10^6 \text{ [J/h]}$

$$\eta = \frac{\dot{E}_o}{\dot{E}_I} \Rightarrow \dot{E}_I \cdot \eta = \dot{E}_o$$

$$5,5 \cdot 10^6 \cdot 0,72 = 3,96 \cdot 10^6 \text{ [J/h]}$$

gammel effekt ut

$$\frac{5,5 \cdot 10^6 \cdot 0,72}{0,82\%}$$

$$0,82\%$$

ny effekt inn

$$\approx 4\,829\,268 \text{ [J/h]}$$

$$b) 3,96 \cdot 10^6 \text{ [J/h]} \cdot \left(\frac{1}{0,82} - \frac{1}{0,72} \right) \cdot 3900 \text{ [h]}$$
$$\approx 26\,158\,536\,58,5 \text{ [J]}$$

c) $\cdot 0,4 \cdot 10^{-6} \text{ [kr/J]}$

$$\approx 10\,46,34146 \text{ [kr]} \text{ spart}$$

(fisienn negativ fordi det er hvor mye lavere)
enn før.